



Implementation and Monitoring MLOps Solution

LECTURE

Monitoring in Machine Learning



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The lecture **Monitoring in Machine Learning** covers ML system monitoring, Lakehouse monitoring practices, capabilities, and the use of Databricks SQL alerts.

Monitoring ML Systems

Continuous logging and review of key component/system metrics

Why?

Used to help diagnose issues before they become **severe** or **costly**

Data to Monitor

Input data (tricky with existing models)
Data in feature stores and vector databases
Human feedback data
Model Outputs

ML Assets to Monitor

Mid-training checkpoints for analysis
Component evaluation metrics
ML system evaluation metrics
Performance/cost details



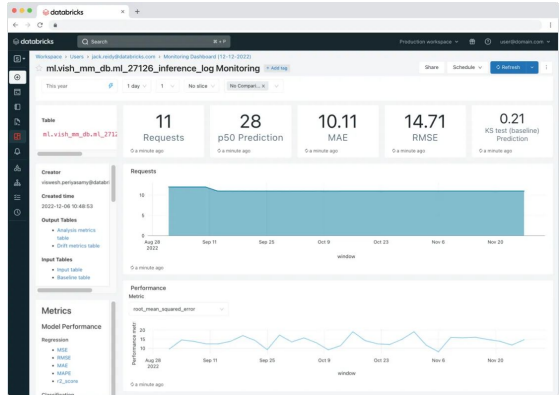
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- **What is monitoring?**
 - Ongoing evaluation and logging of key metrics **used to analyze performance** of a solution
- **Why is it important and helpful?**
 - Can help diagnose issues before they become severe or costly
- What metrics are we including?
 - **Data**
 - Input data (as we can)
 - Data in feature stores and vector database, especially if it updates
 - Human feedback data (trigger?)
 - **ML Systems**
 - Mid-training checkpoint analysis
 - Model Evaluation metrics

Lakehouse Monitoring

Manage, govern, evaluate, and switch models easily

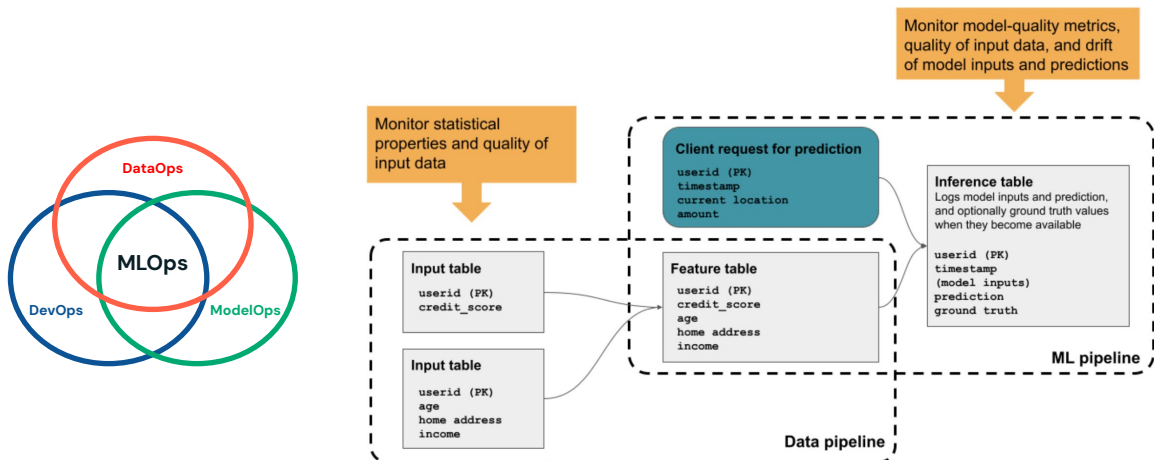
- Monitor data and AI assets
- Centralized and standardized mechanism for monitoring models in production
- Simplified, built-in tool for monitoring mechanisms to diagnose errors, detect drift, etc.
- Allow for the creation of additional custom metrics.
- Alerting to get notified on drift or quality issues.



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Monitoring data and AI assets provides a centralized, standardized approach for tracking models in production. Built-in tools simplify error diagnosis and drift detection, while allowing the creation of custom metrics and alerting systems to notify teams of data quality or performance issues.

What does this look like in practice?



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- **LakeHouse Monitoring Integrates Monitoring Needs in all areas of MLOps for comprehensive management of data and models.**
 - Monitors statistical properties and ensures high-quality input data.
 - **Inference tables** log details of every prediction request and the corresponding model output.
 - i. Includes timestamps and model details to track performance and inputs over time.
 - Tracks model-quality metrics, including drift of model inputs and predictions to maintain model accuracy.
- **Integration with Ground Truth Data**
 - Ground truth values are logged alongside predictions to facilitate continuous model training and validation.
 - Helps in refining model predictions against actual outcomes, enhancing model relevance and accuracy.

Lakehouse Monitoring Capabilities



Monitor Key Metrics

For fields such as Nulls, null %, zeros, zero %, avg, distincts, distinct %, max, min, stdev, median, max/min/avg length, value frequencies, quantiles, row counts



Define (multiple) time granularities

Monitor metrics over time windows, e.g. every day, every 5 minutes, over n weeks



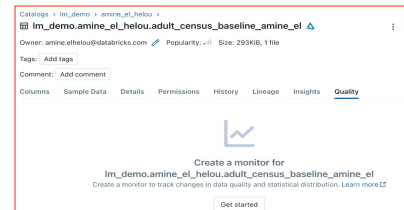
Monitor Data Slices

Slice metrics based on columns or predicates, e.g. state, product_class, "cart_total > 1000"



Monitor Tables, VIEWS and ML Models

Consistent quality & drift monitoring of all your production assets including machine-learning model's fairness & bias



UI (UC Data Explorer)

```
import databricks.lakehouse_monitoring as lm

# Set up monitoring parameters
lm.create_monitor(
    table_name="my_UC_table",
    ...)

# Refresh monitoring metrics
lm.run_refresh("my_table")
```

Python API



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Our API consists of essentially two calls

- An idempotent call `create\_or\_update\_monitor` to set up monitoring on a table (essentially, the config for monitoring).
- A call to recompute the monitoring metrics based on the latest contents of the table.

Eventually the user will not need to make the refresh explicit, and we will provide the functionality of automatic refreshes based on some policy.

For now, the user can include these calls inside their existing pipelines (e.g., call `refresh_metrics()` at the end of the pipeline that writes the table) or schedule a job to run the refresh periodically.

Lakehouse Monitoring Capabilities



Dashboards and Alerts

Auto-generated DB SQL dashboards to visualize metrics & trends, SQL Alerts for notifications



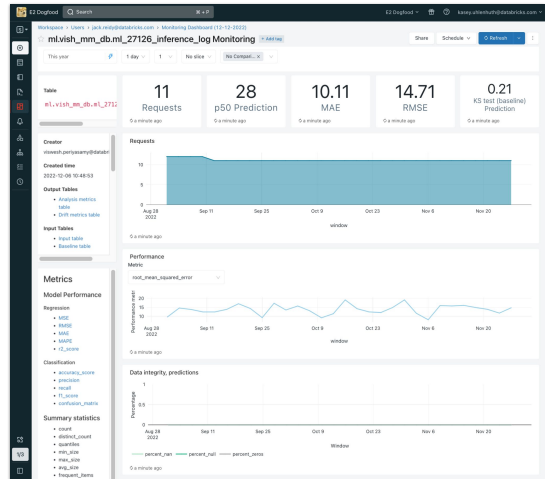
Open Monitoring Results

Monitoring results stored in open format Delta tables to build custom analytics using your favorite BI tool



Simple Operations

Databricks managed compute eliminates infrastructure management and scaling complexity

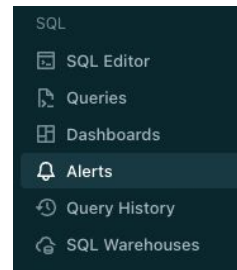


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Lakehouse Monitoring provides auto-generated Databricks SQL dashboards to visualize metrics and trends, along with SQL Alerts for notifications. Monitoring results are stored in open-format Delta tables, allowing the use of any BI tool for custom analytics. Databricks-managed compute ensures simple operations by removing the complexity of infrastructure management and scaling.

What are Databricks SQL alerts?

- Databricks SQL alerts **periodically run** queries, **evaluate defined conditions**, and **send notifications if** a condition is met.
 - **Scheduled Execution:** Automatically runs queries at defined intervals to check specific conditions.
 - **Multi-Channel Notifications:** Receive alerts via Email, Slack, Webhook, MS Teams, PagerDuty, and more.
- Explore the [documentation](#) for in-depth setup and customization options.

A screenshot of the Databricks SQL alert configuration interface. It shows a 'Trigger condition' section with 'Value column' set to 'Total Price', 'Operator' set to '>', and 'Threshold value' set to '1000'. A dropdown menu for 'When alert is triggered' is open, showing options like 'First row', 'Sum', 'Count', 'Count distinct', 'Average', 'Median', 'Min', 'Max', and 'Standard deviation'. The 'Sum' option is selected. There are also sections for 'Notifications' and 'Template'.

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- **Databricks SQL Alerts** = SQL alerts in Databricks automatically execute predefined queries at regular intervals to monitor specific conditions within your data.
 - Alerts assess these conditions and **send notifications if** certain criteria are met
 - Each alert runs independently based on its own schedule, regardless of the query's execution plan, allowing for timely checks and updates.
- **Parameter Usage:** When alerts leverage queries with parameters, the default values set in the SQL editor are used, streamlining the alert process.
- **Notification Process**
 - Alerts can **send notifications through various channels** including Email, Slack, Webhooks, Microsoft Teams, and PagerDuty, facilitating immediate action and response.
- **Alerts Organization:** Alerts can be organized into folders within the Databricks workspace
- **CI/CD Integration:** Alerts can be used to trigger retraining, Workflow Jobs, or CI/CD pipelines
- Explore the [documentation](#) for in-depth setup and customization options.



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